

Generalized Gated Four-Probe Method for Intrinsic Mobility Extraction With Van Der Pauw Structure

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Abstract—Measuring the mobility of thin-film transistors fabricated using newly developed semiconductors is difficult because of various factors, such as contact resistance. In this work, the generalized gated four-probe and van der Pauw structure were combined to precisely measure the carrier mobility of semiconductors. The method is geometry-independent, suitable for gate-dependent mobility in all operation regimes, and the test devices could be easily fabricated. Experimental verifications were performed on specially designed InGaZnO devices with different electrodes and at various temperatures. The developed approach enables simple and precise investigation of the semiconductor's properties via easily fabricated devices. In particular, the technique could be used for newly developed semiconductors that can only be patterned via low-resolution techniques, such as shadow mask patterning.

Index Terms—Field-effect transistor, semiconductor, mobility, gated van der Pauw, generalized gated four-probe.

I. INTRODUCTION

THE realization of high charge carrier mobility is one of the primary motivations for the development of new semiconductors [1]; however, determination of the charge-carrier mobility in field-effect transistors (FETs) or thin-film transistors (TFTs) has proven difficult [2]. Specifically, nonideal contact injections [3] complicate the FET's transfer curves, which leads to erroneous mobility values [4, 5]. A significant amount of work has gone into addressing this issue [3–6]. Previous methods (transfer line method (TLM) [4] and conventional gated four-probe (GFP) [7]) can only be applied in restricted conditions (i.e., in the linear regime $|V_D| \ll |V_G|$) and usually require precise patterning, which is challenging for some newly developed materials [8] that could be easily damaged via conventional photolithography [9, 10]. To this end, van der Pauw (VDP) resistance measurement [11] has been recently adjusted to extract the mobility of semiconductors via the

application of gate bias [12], and the accuracy of this method is suggested to be independent of the device's geometrical dimensions.

However, this method is still limited to the linear regime (i.e., small drain voltage), relies on the assumption of constant carrier mobility, and uses a constant current source. In this work, these limitations were addressed by combining the generalized GFP method [7] with the VDP structure (denoted as GFP-V). This device structure only requires a simple square shape [13], and the measurements are suitable for arbitrary drain voltages (i.e., in all operation regimes) with a normal voltage source. Furthermore, the mobility extraction equation does not require the geometrical parameter of the devices. The results demonstrate that GFP-V measurements can enable accurate extraction of the gate-field dependent carrier mobility at various temperatures, and the results were not affected by metal contacts. In particular, the GFP-V measurement corrected the contact-induced nonideal μ - V_G curves in defective TFT devices and provided precise values of the gate-dependent carrier mobility.

II. DEVICE FABRICATION AND CHARACTERIZATIONS

All the TFT and VDP devices were fabricated on a Si/SiO₂ wafer with heavily doped Si serving as the bottom gate and 100-nm thick SiO_x as the gate dielectric. The InGaZnO (IGZO) semiconductor was adopted as the active layer because of its well-studied properties [14]. The 60-nm-thick IGZO film was sputtered, grown, and defined as a square-shaped island with a side length $L_{IGZO} = 400 \mu\text{m}$. Contact pads were formed with either 100-nm-thick sputter-grown Ti or 80-nm-thick thermally evaporated Au. For the TFTs, source/drain electrodes were patterned wider than the IGZO island to avoid fringe effects (Fig. 2a) [15]. The channel width W was fixed at $400 \mu\text{m}$, and length L was patterned to be 200 or $80 \mu\text{m}$. For the VDP devices (Fig. 2a), four contact pads were patterned via a lift-off process to overlap the corners of the IGZO island in an isosceles triangle shape (side length $l = 80 \mu\text{m}$).

The electrical properties were measured using an Agilent B1500A analyzer under an ambient environment (room-temperature tests) or an N₂ atmosphere (temperature-dependent measurements). For characterization of the VDPs, two electrodes on one side were applied with bias (V_S , V_D) while the other two electrodes probed the potentials, as illustrated in Fig. 1a. The capacitance per unit area C_i is 34.5 nF/cm^2 .

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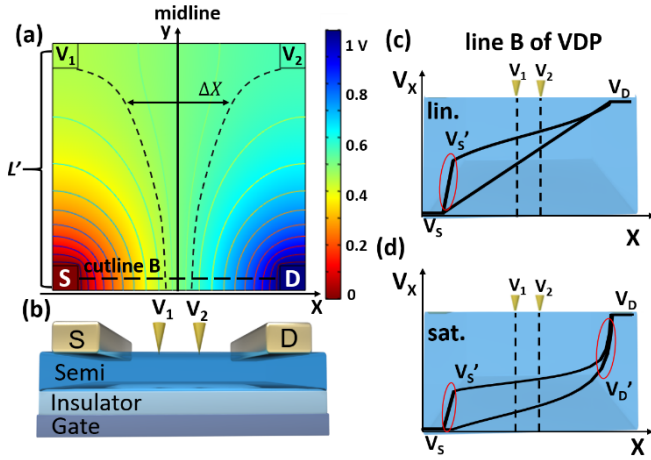


Fig. 1. (a) Simulated potential contours at the top surface of a square-shaped VDP device with the bottom two electrodes applied with bias (V_S , V_D). The black dashed line across the top two electrodes indicates that the operation of VDP is similar to a gated four-probe TFT with two potential probes (V_1 , V_2). Dashed line B is a horizontal cutline of the VDP device, which crosses the S/D electrodes in the transverse direction. (b) Cross-section view of the equivalent TFT. (c) Simulated potential (V_X) along cutline B in the linear regime for ohmic and Schottky contact; (d) is the same for the saturation regime. The V_s near the source contact denotes the potential consumption of the Schottky barrier, and the V_d near the drain contact denotes the potential consumption of the depletion region.

III. RESULTS AND DISCUSSION

For a TFT device, the GFP measurement began by probing the potentials of two points on the channel (V_1 , V_2) equal distance from the midpoint. Experimental observations and device simulations [16] show that the change of the electrostatic potential is generally gradual in the channel except near the contact-depleted regions. Subsequently, the potential V_X and lateral electrical field E_X at the midpoint could be deduced from V_1 and V_2 for all operation regimes by:

$$V_X \cong \frac{(V_1 + V_2)}{2}, \quad E_X \cong \frac{(V_1 - V_2)}{(X_2 - X_1)} = -\frac{\Delta V_X}{\Delta X} \quad (1)$$

Therefore, the carrier concentration n and sheet conductance σ at the midpoint are:

$$n = C_i (V_G - V_{TH} - V_X) / q, \quad \sigma = nq\mu \quad (2)$$

where, V_{TH} is the threshold voltage, and q is the elementary charge. The drain current can be described with the drift current equation:

$$I_D = W\sigma |E_X| = W\sigma \left| \frac{\Delta V_X}{\Delta X} \right| \quad (3)$$

By substituting (1) and (2) into (3) and calculating the differential, the mobility extraction equation for the TFT configuration can be obtained as:

$$\mu_{GFP} = \frac{\Delta X}{WC_i} \frac{\partial I_D}{\partial [(V_G - V_{TH} - V_X) \Delta V_X]} \quad (4)$$

Device simulations have shown that using (1) and (4) to estimate V_X , E_X , and μ is much more accurate than conventional TFT measurements [7]. However, the gated four-probe in the TFT configuration requires finely patterned probes to avoid the channel shunting effect [17].

A simpler structure is a square-shaped VDP device as shown in Fig. 1a, where the charge transport is similar to that in a TFT. Using 3D TCAD simulation, the probed potentials of the VDP devices are verified to indeed be equidistant from the midpoint (Fig. 1c–d) for arbitrary regimes and contacts.

Compared with the regular GFP technique, the accuracy of the mobility extracted from GFP-V is maintained even when using low-resolution patterning techniques.

For the VDP devices, along the midline (see Fig. 1a), the carrier density n is constant because $V_X \cong \frac{(V_1 + V_2)}{2}$ is constant. However, the electrical field E_X evolves with its longitude position, i.e., $E_X(y) \cong -\frac{(V_1 - V_2)}{\Delta X(y)} = -\frac{\Delta V_X}{\Delta X(y)}$, where ΔV_X is constant but $\Delta X(y)$ is dependent on y . Then, the total current is the integral along the midline:

$$I_D = qn\mu \int_0^{L'} |E_X(y)| dy \cong \sigma_m \Delta V_X \int_0^{L'} \frac{dy}{\Delta X(y)} = \sigma_m \Delta V_X A \quad (5)$$

where, σ_m is the midpoint conductance, and A is the geometrical parameter regardless of the operational regime.

For a square-shaped VDP device, the measured current I_D and probed potential difference ΔV_X is dependent on the sheet conductance σ , as described by [11]:

$$2\exp\left(-\pi \frac{\Delta V_X}{I_D} \sigma\right) = 1, \quad (6)$$

In the linear regime, σ is uniform over the whole device. Hence, if we substitute (5) into (6), we get: $A = \frac{\pi}{\ln 2}$. Note that the region described by parameter A is bordered by the potential contour of V_1 and V_2 (see in Fig. 1a) and far from the depletion region around the contacts. Additionally, the potential change within this region is gradual regardless of the operational regimes. Therefore, the constant $A = \frac{\pi}{\ln 2}$ is applicable for all regimes above the threshold, and the current can be expressed as:

$$I_D = \frac{\pi}{\ln 2} \sigma_m \Delta V_X = \frac{\pi}{\ln 2} C_i \mu \left(V_G - V_{TH} - \frac{V_X}{2} \right) \Delta V_X \quad (7)$$

Finally, by calculating the differential of the current expression, the mobility for all regimes can be extracted as:

$$\mu_{GFP-V} = \frac{\ln 2}{\pi C_i} \frac{\partial I_D}{\partial [(V_G - V_{TH} - V_X) \Delta V_X]} \quad (8)$$

For the experiments, the devices were first fabricated in a square-shaped VDP structure with Ti electrodes and characterized via the GFP-V method. For direct comparison, *a*-IGZO TFTs ($W/L = 2$ or 5) were built on the same substrate and characterized via the conventional FET method [2]. During the I - V tests, V_G was swept from -15 to 25 V (0.25 V steps), while V_D was fixed at 20 V (saturation regime). As seen in Fig. 2a, both the TFTs and VDP show similar threshold voltage V_{TH} ($4.3 \sim 5.4$ V). The probed potentials (V_1 and V_2) are linearly dependent on V_G (Fig. 2b). As a result, the calculated carrier concentration $n_{GFP-V} = C_i (V_G - V_{TH} - V_X) / q$ increases linearly with V_G (Fig. 2c), which is consistent with the classic parallel capacitor assumption. In Fig. 2d, the μ_{FET} extracted from the TFTs with $L = 80$ and $200 \mu\text{m}$ coincide with each other, which indicates ideally ohmic contacts between IGZO and Ti. Moreover, the good agreement between μ_{FET} and μ_{GFP-V} (Fig. 2d) confirms the accuracy of the GFP-V method for probing semiconductor mobility in devices with ohmic contacts.

Next, devices with Au electrodes that formed Schottky contact were built and studied. The extracted V_{TH} are about $4.8 \sim 5.0$ V (Fig. 3a). The deviation between the μ_{FET} extracted from the TFTs with different L indicates that the IGZO-Au contact is non-ohmic and distorts the potential evolution

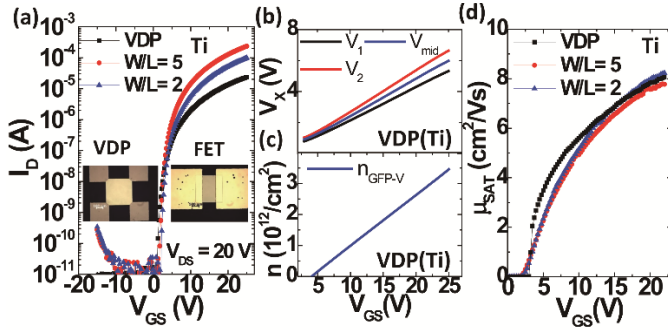


Fig. 2. (a) Transfer curves for VDP and FET devices built with Ti contacts. The inserts are the corresponding optical images. (b) Measured potential V_X of the probes and the calculated mid-channel potential V_{MID} as a function of V_G . (c) Calculated carrier concentration n_{GFP-V} . (d) Extracted μ_{FET} and μ_{GFP-V} as a function of V_G .

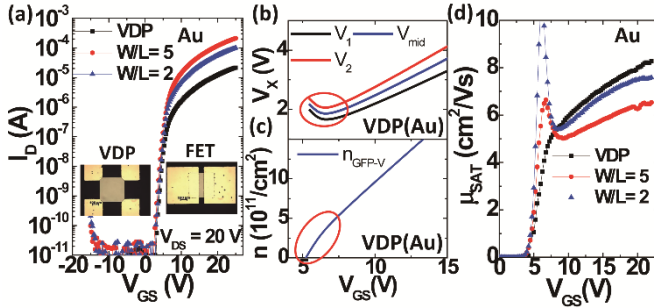


Fig. 3. (a) Transfer curves for VDP and FETs built with Au contact. The inserts are the corresponding optical images. (b) The measured potential of the probes V_X and the calculated mid-channel potential V_{MID} as a function of V_G . (c) Calculated carrier concentration n_{GFP-V} . The red circle indicates its deviation from the ideal accumulation model. (d) Extracted μ_{FET} and μ_{GFP-V} as a function of V_G . The peak value for the TFT ($W/L = 2$) is beyond the display panel.

(Fig. 3d). Unlike devices with a Ti contact, the probed V_X here show non-monotonical change with V_G (Fig. 3b), which confirms the presence of a Schottky barrier. In the FET measurements, the values of μ_{FET} extracted from TFTs abruptly rise to a peak and decrease drastically when V_G is scanned from 5 to 8 V (see in Fig. 3d). This is a typical behavior caused by the transition from the Schottky barrier transistor mode to the TFT mode and leads to significantly overestimated mobility values [5]. In the GFP-V measurements, the extracted carrier concentration n_{GFP-V} significantly deviates from the ideal parallel capacitor assumption (Fig. 3c). The overestimated carrier mobility can be attributed to such nonlinear accumulation. In contrast, the μ_{GFP-V} extracted from the VDP device increases monotonically with V_G (Fig. 3d) and is consistent with that of a TFT with Ti contacts. These results validates the applicability and accuracy of the GFP-V method in revealing the intrinsic mobility of materials with arbitrary contacts.

Furthermore, we verified the proposed method in the measurements under various temperatures (~ 213 – 313 K, at 20 K steps), as shown in Fig. 4. Usually, thermally activated charge transport in IGZO TFTs would follow the Arrhenius equation: $G = G_0 \exp(-E_A/kT)$, where $G = I_D/V_D$ is the channel conductance, k is the Boltzmann constant, T is the temperature, and E_A is the activation energy [18]. The data for the IGZO-Ti TFT ($W/L = 5$) fits well with the Arrhenius equation (Fig. 4a). However, the fitting becomes poor for the IGZO-Au TFTs (Fig. 4d), which is because of the Schottky contacts.

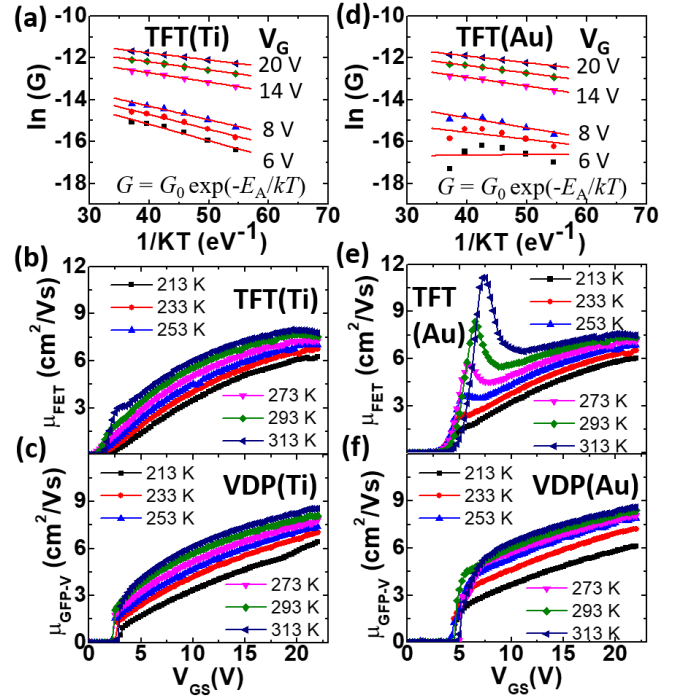


Fig. 4. (a) Temperature-dependent measurement (213 ~ 313 K, 20 K a step), extracted, (b) μ_{FET} and (c) μ_{GFP-V} of devices with Ti contact. (d-f) are the same for IGZO-Au devices.

The μ_{FET} and μ_{GFP-V} of IGZO-Ti devices are mutually consistent and increase monotonically with V_G , as illustrated in Fig. 4b–c. However, for IGZO-Au TFTs, the μ_{FET} shows sharp peaks against V_G (Fig. 4e), and the results suggest that overestimation because of the transition in the operational modes becomes more severe as the temperature increases. By contrast, the μ_{GFP-V} extracted from the VDP device with Au contacts increases monotonically with V_G (Fig. 4f) and is in good agreement with that of the IGZO-Ti devices (Fig. 4b–c) at various temperatures. The comparative studies also show that the GFP-V measurements could provide accurate measurements of the carrier mobility in defective devices and detections of the intrinsic transport characteristics in temperature-dependent investigations.

IV. CONCLUSION

In conclusion, after combining the generalized GFP technique with the VDP structure, the resultant GFP-V measurements were able to exclude geometry dependencies, correct for the contact effects, and identify the intrinsic gate-dependent mobility of semiconductors. The accuracy of this method was verified on IGZO devices with ohmic or Schottky contact and at various temperatures. Given its general applicability and good accuracy, we suggest that the GFP-V method could be a useful tool to characterize the electrical properties of newly developed semiconductors, which encounter difficulties related to complicated patterning and exhibit gate-field-dependent transport properties. Hence, the demonstrated method could pave the way for understanding the potential of newly developed semiconductor materials in building functional circuit units.

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